

THE ζ FUNCTION ANSWER TO PARITY VIOLATION IN THREE-DIMENSIONAL GAUGE THEORIES

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We study parity violation in $(2 + 1)$ -dimensional gauge theories coupled to massive fermions. Using the ζ function regularization approach we evaluate the ground state fermion current in an arbitrary gauge field background, showing that it gets two different contributions which violate parity invariance and induce a Chern–Simons term in the gauge field effective action. One is related to the well-known *classical* parity breaking produced by a fermion mass term in three dimensions; the other, already present for massless fermions, is related to peculiarities of gauge-invariant regularization in odd-dimensional spaces.

1. Introduction

Gauge theories in three-dimensional space–time exhibit a variety of phenomena that are of interest not only in quantum field theory¹ but also in condensed matter physics.²

An important feature of three-dimensional kinematics concerns the possibility of giving a (topological) mass to the vector field by including an unconventional term in the gauge field Lagrangian:^{3–5} the Chern–Simons (CS) secondary characteristic.

The CS term, of topological origin, violates both P and T invariances. Since the same happens for the mass term for a (two-component) Dirac spinor in three-dimensional space–time, it is natural to expect an interesting interplay between the two masses in three-dimensional gauge theories coupled to fermions. Indeed, in Refs. 3, 4, 6 and 7 it was shown that if any of the two mass terms is included in the Lagrangian, the other is then induced by radiative corrections.

Now, in Refs. 6 and 7, it was also shown that even massless fermions, when coupled to gauge fields, generate a CS term. Originally this effect was thought to be a consequence of the introduction of a fermion mass term within the Pauli–Villars regularization procedure.⁷ However, the occurrence of the CS term was confirmed in Ref. 8 (hereafter referred to as I) using the ζ function approach, *where no regulating*

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mass term is added at any stage of the calculations. In fact, the violation of parity in odd dimensions (and, consequently, the generation of a CS term) is analogous to the nonconservation of the axial current in even dimensions: the imposition of gauge invariance produces in both cases an anomaly for a symmetry (parity, chiral symmetry) of the original action for *any* sensible regularization.

The issue of parity violation in three dimensions is relevant in different contexts. In particular, discrepancies on overall parity breaking have arisen in the analysis of the $2 + 1$ Thirring model,^{9–12} this leading to contradictory results concerning dynamical mass generation for fermions (see Ref. 13 for a treatment of the Thirring model using the techniques described in this paper). Since $(3 + 1)$ -dimensional field theories become effectively three-dimensional in the high temperature limit, the problem has applications in four-dimensional physics too. Also, in condensed matter physics, the appearance of a CS term through parity anomaly provides an adequate ground for testing the physics of anyons in three-dimensional fermionic systems.² In this respect, generation of the CS term through the fermion effective action has direct consequences for the thermodynamic properties of the system.¹⁴

Since the generation of the CS term is a consequence of regularizing the three-dimensional fermionic ground state current, a careful analysis of the regularization prescription is needed in order to decide whether or not parity violation occurs. We address ourselves to this issue in the present work, extending previous results that we have already presented in I.

Firstly, we give a mathematically rigorous scheme leading to the definition of the path integral fermionic measure for a three-dimensional gauge theory. This is achieved through a careful treatment of the fermionic determinant (a not-well-defined object for the unbounded Dirac operator) using the results of Seeley¹⁵ on complex powers of elliptic operators.

Secondly, we give the recipe for computing fermionic ground state currents. Since at this stage we already dispose of a finite expression for the fermionic partition function $\mathcal{Z}$ within the ζ function approach, this recipe reduces to giving explicit formulae for obtaining vacuum expectation values by using the differentiability properties of the ζ function.¹⁶

Finally, we apply our approach to the analysis of massive fermions, thus completing the work initiated in I on massless fermions. We think that the present analysis clarifies the origin of the parity anomaly and, in particular, the appearance of two contributions to parity violation, one originating in peculiarities of calculations in odd-dimensional spaces, the other arising from the parity violation produced, already at the classical level, by the fermion mass in three dimensions. The results we present, together with current investigations like those in Refs. 17–20, show that the ζ function method provides a powerful tool for studying problems in which regularization prescriptions require careful handling.

In respect of massless fermions, the original calculation of the ground state fermion current $j_\mu[A]$ in a constant magnetic background $*F_\mu$, $*F_\mu = i\epsilon_{\mu\nu\alpha}\partial_\nu A_\alpha$

was presented in Ref. 7 (we are working here in Euclidean space). As an example, the perturbative calculation using Pauli–Villars regularization yields for the Abelian case, in the one loop approximation,⁷

$$j_\mu[A] = \frac{m}{|m|} \frac{e}{4\pi} {}^*F_\mu, \quad (1)$$

where e is the gauge coupling constant and m is a fermionic mass to be put to zero at the end of the calculations. This current is identically conserved, but it violates parity conservation since ${}^*F_\mu$ is a pseudovector.

From this current the effective action $\Gamma[A]$ can be computed, using the relation

$$\frac{\delta\Gamma[A]}{\delta A_\mu} = e j_\mu[A]. \quad (2)$$

From Eqs. (1) and (2) we see that the effective action, apart from parity-conserving contributions, contains a CS term,

$$\Gamma[A]_{\text{Pauli-Villars}} = \frac{m}{|m|} \frac{e^2}{2} S_{\text{CS}}[A] + S_{\text{PC}}[A], \quad (3)$$

where $S_{\text{CS}}[A]$ is the (Abelian) Chern–Simons action,

$$S_{\text{CS}}[A] = \frac{1}{4\pi} \int d^3x A_\mu {}^*F_\mu, \quad (4)$$

and $S_{\text{PC}}[A]$ are parity-conserving terms. We have explicitly indicated in Γ that this result was obtained in Ref. 7 within the Pauli–Villars regularization scheme.

Of course, this result can be trivially generalized to the case of N flavors. Instead of (3), one has, when there are N fermion species,

$$\Gamma[A]_{\text{Pauli-Villars}} = \sum_{i=1}^N \frac{m_i}{|m_i|} \frac{e^2}{2} S_{\text{CS}}[A] + S_{\text{PC}}[A, N]. \quad (5)$$

As in the $N = 1$ case, one has to put the fermion masses $m_i = 0$ to recover the massless case. Within this approach, one could evidently choose the signs of m_i in such a way that, for even N , the overall CS contribution to Γ would cancel out so that overall parity would not be violated. Now, using the ζ function approach, we have proven in I, without the necessity of introducing a mass parameter, that instead of (3) one directly obtains

$$\Gamma[A]_\zeta = \pm \frac{e^2}{2} S_{\text{CS}}[A] + S_{\text{PC}}[A]. \quad (6)$$

As will become clear in the next section, within the ζ function approach, the sign ambiguity in (6) can be traced back to the choice of an integration path Ω in the complex plane, necessary for defining the complex powers of the Dirac operator.

In odd-dimensional space-times, the choice of Ω in the upper (lower) half plane yields to a positive (negative) sign in (6). Then, once a definite integration path is chosen, consistency in the definition of complex powers implies that, when N species are present, all fermions contribute with the same sign so that, instead of Eq. (5), one gets

$$\Gamma[A]_{\zeta} = \pm N \frac{e^2}{2} S_{\text{CS}}[A] + S_{\text{PC}}[A, N] \quad (7)$$

and overall parity is always violated.

Ending with a summary of the massless case analysis, one should stress that the ζ function approach allows the calculation of the parity-violating contribution to $j_{\mu}[A]$ for *arbitrary* A_{μ} (and not necessarily for one leading to a constant magnetic field) in an *exact* form. As explained above, it was shown in Ref. 7 that the parity violation contribution was induced from one-fermion loops and arguments were given to discard higher loop corrections. Now, we have proven in I that the CS contribution present in (6) is the *only* parity-violating contribution one has to expect for three-dimensional massless fermions since, as in the case of the chiral anomaly, the ζ function approach gives the exact anomalous contribution.

Let us come now to the massive fermion case. The problem was already discussed in Ref. 3, to the one-loop order and in a constant magnetic background, using Pauli-Villars regularization, and clearly explained in Ref. 1. The conclusion was that when massive fermions couple to the gauge field, a CS term is induced by fermion radiative corrections. The fact that a fermion mass term in three dimensions violates P and T was considered at the origin of this result. The explicit form for the effective action in the massive case was calculated in Ref. 7. The result coincides with (3) except that in this case m is the physical fermion mass. Again, the conclusion was dependent on how to regularize divergent objects.^{3,22,23} It is then worthwhile to analyze the problem of massive fermions using the ζ function approach developed in I. We here undertake such a program, explaining in Sec. 2 the mathematical tools needed for the computation of ground state fermion currents and leaving for Sec. 3 the explicit calculation of j_{μ} . A summary of results and discussions is given in Sec. 4.

2. How to Compute Currents

We consider the Abelian case giving the basic formulae to be used in the next section.

The partition function for a massive Dirac fermion doublet in $d = 3$ Euclidean space-time dimensions is

$$Z[A_{\mu}] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left(- \int \bar{\psi} D[A] \psi d^3x \right), \quad (8)$$

where A_{μ} is a background vector field and the Dirac operator $D[A]$ is given by^a

^aOur conventions for γ matrices are $\gamma_{\mu} = \sigma_{\mu}$, where σ_{μ} ($\mu = 1, 2, 3$) are the Pauli matrices.

$$D[A] = \gamma_\mu (i\partial_\mu + eA_\mu) + im. \quad (9)$$

In (8), $\mathcal{D}\bar{\psi}\mathcal{D}\psi$ is some fermionic measure to be defined below. The results we shall describe are valid for any elliptic operator L , not necessarily Hermitian, defined on a compact manifold without boundary.^{16,21} We shall assume, however, that they are also valid for R^3 .

The ground state current $j_\mu[A]$ in the presence of the background field,

$$j_\mu[A] = \langle \bar{\psi}(x)\gamma_\mu\psi(x) \rangle_A, \quad (10)$$

can be calculated from $\mathcal{Z}[A]$ as

$$j_\mu[A] = -\frac{1}{e} \frac{\delta}{\delta A_\mu(x)} \log \mathcal{Z}[A]. \quad (11)$$

Of course, since the Dirac operator is unbounded, $\mathcal{Z}[A]$ needs a regularization,

$$\mathcal{Z}_{\text{reg}}[A] \equiv \det D[A]|_{\text{reg}}. \quad (12)$$

We shall adopt in this work the ζ function regularization method which automatically ensures *local gauge invariance*. We thus define as usual

$$\zeta(D[A], s) = \text{Tr}(D^{-s}[A]), \quad (13)$$

since $D[A]$ is invertible. From (13) we can define

$$\mathcal{Z}_{\text{reg}}[A] = \exp \left[- \frac{d}{ds} \zeta(D[A], s) \right] \Big|_{s=0}, \quad (14)$$

where the right hand side should be computed for complex s (with a large enough real part) and then analytically extended to $s = 0$. This definition amounts in practice to the definition of the fermionic path integral measure. It has been proven in Ref. 16 that $\mathcal{Z}[A]$ written as in Eq. (14) is differentiable; then, one can easily obtain from Eqs. (11) and (14) a regularized expression for $j_\mu[A]$. Indeed, one can write

$$\int \frac{\delta}{\delta A_\mu(x)} \log \mathcal{Z}_{\text{reg}}[A] a_\mu(x) d^3x = \frac{d}{dt} \log \mathcal{Z}_{\text{reg}}[A_\mu + ta_\mu] \Big|_{t=0}. \quad (15)$$

(Here a_μ indicates a direction in A_μ functional space along which the derivative is taken.) Now, from Eq. (14) one has

$$\int \frac{\delta}{\delta A_\mu(x)} \log \mathcal{Z}_{\text{reg}}[A] a_\mu(x) d^3x = \frac{d}{dt} \left(- \frac{d}{ds} \zeta(D[A_\mu + ta_\mu], s) \Big|_{s=0} \right) \Big|_{t=0} \quad (16)$$

or, using Eq. (13) and interchanging the order of the derivatives,

$$\int \frac{\delta}{\delta A_\mu(x)} \log \mathcal{Z}_{\text{reg}}[A] a_\mu(x) d^3x = \frac{d}{ds} [s \text{Tr}(D^{-s-1}[A] e\gamma_\mu a_\mu)] \Big|_{s=0}. \quad (17)$$

Notice that differentiability with respect to t proven in Ref. 16 together with analyticity in s allow the interchange in the order of derivatives. In Eqs. (13) and (17), “Tr” stands for the operator trace, which includes the matrix trace “tr” and an integration over the space–time of the diagonal of the operator kernel. If we denote by $K_s(x, y; D[A])$ the kernel of the operator $D^s[A]$,

$$D^s[A]f(x) = \int d^3y K_s(x, y; D[A])f(y), \quad (18)$$

we can read from Eq. (17)

$$\frac{\delta}{\delta A_\mu(x)} \log \mathcal{Z}_{\text{reg}}[A] = e \frac{d}{ds} \left\{ s \text{Tr}[\gamma_\mu K_{-s-1}(x, x; D[A])] \right\} \Big|_{s=0}. \quad (19)$$

Then, from Eqs. (11)–(19), we have

$$j_\mu^{\text{reg}}[A] = - \frac{d}{ds} \left\{ s \text{Tr}[\gamma_\mu K_{-s-1}(x, x; D[A])] \right\} \Big|_{s=0}. \quad (20)$$

This is the key formula we shall employ to compute fermionic currents in the ζ function regularization scheme.

Note that $K_{-s-1}(x, y; D[A])|_{s=0}$ is nothing but the Green function $G(x, y)$ of the operator $D[A]$ when $x \neq y$. Correspondingly, it is singular on the diagonal $x = y$. Now, for any elliptic operator L of positive order r in a space of dimension n , the kernel of L^s is a continuous function of x, y for $\text{Re}(s) < -n/m$ that admits an analytic extension to the whole complex plane s if $x \neq y$. On the diagonal $x = y$ it is a meromorphic function of s that has at most simple poles at $s = (-n + j)/r$, $j = 0, 1, \dots$.¹⁵ For the Dirac operator $D[A]$ in three-dimensional Euclidean space–time, $K_{-s-1}(x, x; D[A])$ has a simple pole at $s = 0$. It is then clear that Eq. (20) gives j_μ^{reg} as the finite part of the Laurent series for $-\text{tr}[\gamma_\mu K_{-s-1}(x, x; D[A])]$ at $s = 0$, thus providing an operative (finite) formula for $-\text{tr}[\gamma_\mu G(x, x)]$:

$$- \frac{d}{ds} \left\{ s \text{tr}[\gamma_\mu K_{-s-1}(x, x; D[A])] \right\} \Big|_{s=0} = -\text{tr}[\gamma_\mu G^{\text{reg}}(x, x)]. \quad (21)$$

The right hand side of (21) gives the ζ function analog of the usual expression that can be found in the literature²⁴ for the ground state current, with the regularization procedure specified in the present approach by the left hand side.

We see at this point that for computing $j_\mu^{\text{reg}}[A]$ we need an explicit formula for the kernel $K_{-s-1}(x, y; D[A])$ so as to evaluate the r.h.s. of Eq. (20). For the sake of clarity and consistency, let us present some of the results of Seeley¹⁵ on complex powers of elliptic operators in the restricted form needed for studying the Dirac operator.

The complex powers of the elliptic differential operator $D[A]$ (of order 1 in a space of dimension $n = 3$) are best described in terms of the symbol of the operator

and the symbol of its resolvent $(D[A] - \lambda)^{-1}$. The symbol of $D[A]$ is a polynomial $\sigma(D[A])$ in a vector ξ_μ , which can be thought of as the Fourier variable associated with x_μ . It is obtained from $D[A]$ by replacing $-i\partial_\mu \rightarrow \xi_\mu$ and takes the form

$$\sigma(D[A])(x, \xi) = -\gamma_\mu \xi_\mu + e\gamma_\mu A_\mu(x) + im \equiv a_1(x, \xi) + a_0(x, \xi), \quad (22)$$

where $a_j(x, \xi)$ ($j = 0, 1$) are homogeneous functions of degree j in ξ_μ .

The resolvent $(D[A] - \lambda)^{-1}$ of the differential operator $D[A]$ is a pseudodifferential operator.²⁵ Its symbol is a generalization of the definition above and can be properly approximated by

$$\sigma((D[A] - \lambda)^{-1})(x, \xi) = \sum_{j=0}^{\infty} b_{-1-j}(x, \xi; \lambda). \quad (23)$$

Here $b_{-1-j}(x, \xi; \lambda)$ are homogeneous functions of degree $-1-j$ in ξ_μ and λ . These coefficients can be recursively evaluated for each order in ξ_μ from the relation^b

$$\sigma((D[A] - \lambda)^{-1}) \circ \sigma((D[A] - \lambda)) = I, \quad (24)$$

provided one takes λ order ξ , so that it combines with the top term a_1 of the symbol of $D[A]$. The recursive formula reads

$$b_{-1}(a_1 - \lambda) = I, \quad (25)$$

$$b_{-1-l}(a_1 - \lambda) + \sum_{j=0}^{l-1} \sum_{\alpha=0}^{l-j} \frac{1}{\alpha!} \left(\frac{\partial}{\partial \xi} \right)^\alpha b_{-1-j} \left(-i \frac{\partial}{\partial x} \right)^\alpha a_{1-l+\alpha+j} = 0, \quad l = 1, 2, \dots,$$

where $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ is a vector of nonnegative integers, $\alpha = \sum_{i=1}^3 \alpha_i$ and $\left(\frac{\partial}{\partial \xi} \right)^\alpha = \prod_{\mu=1}^3 \left(\frac{\partial}{\partial \xi_\mu} \right)^{\alpha_\mu}$, $\left(\frac{\partial}{\partial x} \right)^\alpha = \prod_{\mu=1}^3 \left(\frac{\partial}{\partial x_\mu} \right)^{\alpha_\mu}$.

From Eqs. (22) and (25) the coefficients of the symbol of the resolvent can be evaluated for the Dirac operator. The first three of them are

$$b_{-1}(x, \xi; \lambda) = \frac{1}{\lambda^2 - \xi^2} (\not{x} - \lambda), \quad (26)$$

$$b_{-2}(x, \xi; \lambda) = -\frac{1}{(\lambda^2 - \xi^2)^2} (\not{x} - \lambda)(e\not{A} + im)(\not{x} - \lambda), \quad (27)$$

$$b_{-3}(x, \xi; \lambda) = \frac{1}{(\lambda^2 - \xi^2)^3} (\not{x} - \lambda)(e\not{A} + im)(\not{x} - \lambda)(e\not{A} + im)(\not{x} - \lambda) \\ + ie \frac{\partial_\mu A_\nu}{(\lambda^2 - \xi^2)^3} [2\xi_\mu (\not{x} - \lambda) \gamma_\nu (\not{x} - \lambda) + (\lambda^2 - \xi^2) \gamma_\mu \gamma_\nu (\not{x} - \lambda)]. \quad (28)$$

^bThe product of symbols is defined in such a way that that it corresponds to the symbol of the composition of operators.²⁵

Following Seeley,¹⁵ the complex powers $D^s[A]$ can be defined as a generalization of Cauchy integral representation of z^s ,

$$z^s = \frac{i}{2\pi} \oint_C \frac{\lambda^s}{z - \lambda} d\lambda. \tag{29}$$

Here, the contour C must avoid a loglike cut in the complex plane λ and encircle the pole of the integrand. For the operator $D[A]$, one must take a contour Ω which encircles the whole spectrum of $D[A]$ and avoids a ray going from the origin to infinity in a direction such that no eigenvalue of a_1 lies on it (such a ray is called a ray of minimal growth or an Agmon ray). For $\operatorname{Re}(s) < 0$ one can take Ω as a curve beginning at ∞ , passing along the ray of minimal growth to a small circle about the origin, then clockwise around the circle, and back to ∞ along the ray. The ray along which the curve Ω is defined cannot coincide with the real axis, since a_1 eigenvalues are real. Two possible Ω curves are illustrated in Fig. 1.

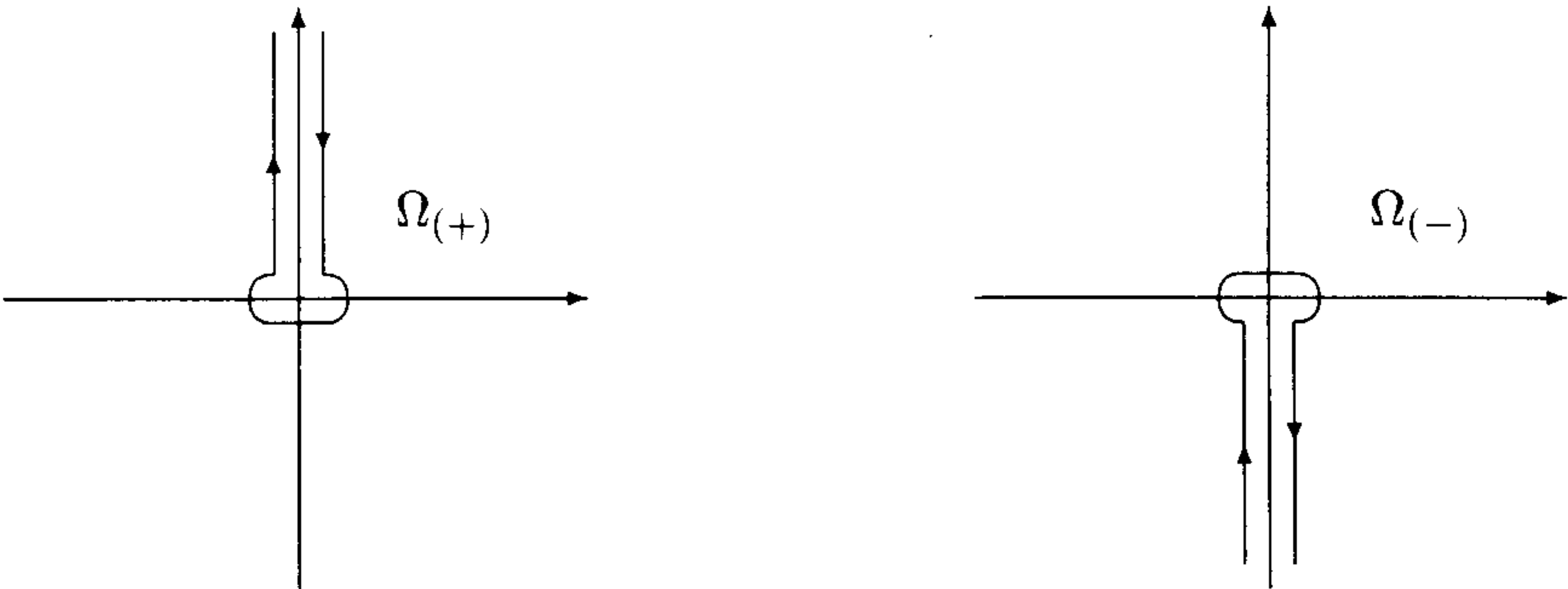


Fig. 1. Contours Ω , for defining complex powers of $D[A]$.

Then, for $\operatorname{Re}(s) < 0$, one defines

$$D^s[A] = \frac{i}{2\pi} \int_{\Omega} \lambda^s (D[A] - \lambda)^{-1} d\lambda. \tag{30}$$

This definition is analytically extended to the whole complex plane through multiplication by integer powers of $D[A]$:

$$D^s[A] = D^k[A] D^{s-k}[A], \quad k \text{ integer}, \quad -1 \leq \operatorname{Re}(s) - k < 0. \tag{31}$$

The corresponding symbol is constructed using Eq. (23) and reads

$$\sigma(D^s[A])(x, \xi) = \sum_{j=0}^{\infty} \frac{i}{2\pi} \int_{\Omega} \lambda^s b_{-1-j}(x, \xi; \lambda) d\lambda \equiv \sum_{j=0}^{\infty} C_j(x, \xi; s), \tag{32}$$

where

$$C_j(x, \xi; s) = \frac{i}{2\pi} \int_{\Omega} \lambda^s b_{-1-j}(x, \xi; \lambda) d\lambda \tag{33}$$

are homogeneous functions of (complex) degree $s - j$. It is worth noting that, while Eq. (30) is valid for $\text{Re}(s) < 0$, the relation between symbols (32) is valid for any value of s .

Using the tools described above we are now in a position to write the kernel $K_{-s-1}(x, \xi; D[A])$ for the operator $D^{-s-1}[A]$, necessary for evaluating the ground state current [Eq. (20)]. Indeed, one of the main points in Seeley's work¹⁵ is that one can approximate kernel K_s through Fourier transforms of the C_j coefficients:

$$K_s(x, y; D[A]) = \sum_{j=0}^N \int \frac{d\xi}{(2\pi)^n} C_j(x, \xi; s) e^{i(x-y)\cdot\xi} + R(x, y; s). \quad (34)$$

Here $R(x, y; s)$ includes terms which will not contribute to the derivative in Eq. (20) at $s = 0$ if N is taken sufficiently large. Though the integral in each term converges only for $\text{Re}(s) < (-n + j)$ (n being the dimension of space-time; here $n=3$), this expression can be analytically extended to arbitrary s for $x \neq y$. On the diagonal $x = y$, $K_s(x, x; D[A])$ is extended to a meromorphic function with at most simple poles at $s = (-n + k)$, $k = 0, 1, \dots$, one arising from each term included in the sum in Eq. (34).

In order to compute j_μ^{reg} in Eq. (20) one uses Eq. (34) to evaluate the finite part of the simple pole of $K_s(x, x; D[A])$ at $s = -1$. To this end, we shall use the following proposition (proven in I and adapted here to the case of the Dirac operator):

Proposition. For the elliptic invertible first order operator $D[A]$ on a three-dimensional compact manifold M without boundary, the following identity holds:

$$\begin{aligned} & \left. \frac{d}{ds} (s K_{s-1}(x, x; D[A])) \right|_{s=0} \\ &= \lim_{y \rightarrow x} \left[G(x, y) - \int \frac{d^3\xi}{(2\pi)^3} C_1(x, \xi/|\xi|; -1) |\xi|^{-2} e^{i\xi \cdot (x-y)} \right. \\ & \quad \left. - \int \frac{d^3\xi}{(2\pi)^3} C_0(x, \xi/|\xi|; -1) |\xi|^{-1} e^{i\xi \cdot (x-y)} - \int_{|\xi| \geq 1} \frac{d^3\xi}{(2\pi)^3} C_2(x, \xi; -1) e^{i\xi \cdot (x-y)} \right] \\ & \quad - \int_{|\xi|=1} \left. \frac{d}{ds} C_2(x, \xi; s) \right|_{s=-1} \frac{d^2\xi}{(2\pi)^3}, \end{aligned} \quad (35)$$

where $G(x, y)$ is the Green function of $D[A]$. (The interested reader can find the proof in I.)

The last term between brackets in Eq. (35) can be conveniently rewritten as²⁶ (see I)

$$\int_{|\xi| \geq 1} \frac{d^3\xi}{(2\pi)^3} C_2(x, \xi; -1) e^{i\xi \cdot (x-y)} = h_0(x, x-y) + M(x) [\log |x-y| + C(x-y)], \quad (36)$$

where $h_0(x, z)$ is a homogeneous function of degree 0 defined as

$$h_0(x, z) = \int \frac{d^3\xi}{(2\pi)^3} \text{PV}[C_2(x, \xi/|\xi|; -1) - M(x)]|\xi|^{-3} e^{i\xi \cdot z}, \quad (37)$$

where $M(x)$ is defined as

$$M(x) = \frac{1}{\omega} \int_{|\xi|=1} C_2(x, \xi; -1) d^2\xi, \quad (38)$$

and finally

$$\omega[\log|z| + C(z)] = - \int_{|\xi| \geq 1} \frac{d^3\xi}{(2\pi)^3} |\xi|^{-3} e^{-i\xi \cdot z}, \quad (39)$$

where $C(z)$ is a regular function in the neighborhood of $z = 0$. All Fourier transforms are taken in the sense of distributions and PV means principal value.

The interpretation of Eq. (35) is as follows. The last three terms between brackets subtract the singularities of $G(x, y)$ on the diagonal $x = y$. Concerning the last term of Eq. (35), it is a *local* regular term generated through the regularization procedure that adds to the regular part of $G(x, x)$. In view of this interpretation we rewrite the result in Eq. (35) in the form

$$\left. \frac{d}{ds} (sK_{s-1}(x, x; D[A])) \right|_{s=0} = G^{\text{reg}}(x, x) = G^{(\text{subt})}(x) + G^{(\text{local})}(x), \quad (40)$$

where

$$\begin{aligned} G^{(\text{subt})}(x) &= \lim_{y \rightarrow x} \left\{ G(x, y) - \int \frac{d^3\xi}{(2\pi)^3} C_1(x, \xi/|\xi|; -1) |\xi|^{-2} e^{i\xi \cdot (x-y)} \right. \\ &\quad - \int \frac{d^3\xi}{(2\pi)^3} C_0(x, \xi/|\xi|; -1) |\xi|^{-1} e^{i\xi \cdot (x-y)} \\ &\quad \left. - \int_{|\xi| \geq 1} \frac{d^3\xi}{(2\pi)^3} C_2(x, \xi; -1) e^{i\xi \cdot (x-y)} \right\}, \end{aligned} \quad (41)$$

$$G^{(\text{local})}(x) = - \int_{|\xi|=1} \frac{d}{ds} C_2(x, \xi; s) \Big|_{s=-1} \frac{d^2\xi}{(2\pi)^3}. \quad (42)$$

In spite of its apparent complexity, the result in Eq. (35) can be used with great simplicity for evaluating ground state currents for *arbitrary gauge background fields*. This will be seen in the next section, where we shall employ Eqs. (20) and (35) to compute the parity-violating terms of $j_\mu^{\text{reg}}[A]$.

3. The Current for Massive Fermions

The formula (35) is at the root of the ζ function regularization prescription for the calculation of ground state fermion currents. By the use of this approach, the parity-violating contribution to the ground state current was evaluated in I for the case of massless fermions. In this section the analysis is extended to the case of massive fermions with partition function given by Eq. (8).

Let us start by understanding why parity-violating terms should be expected in a $2 + 1$ fermionic theory. First, note that a three-dimensional fermion mass term constructed from two-component spinors violates parity and time inversion. Indeed, under parity transformation the vector and fermion fields behave as

$$\begin{aligned}\mathcal{P}A_0(x)\mathcal{P}^{-1} &= A_0(x'), & \mathcal{P}A_1(x)\mathcal{P}^{-1} &= -A_1(x'), \\ \mathcal{P}A_2(x)\mathcal{P}^{-1} &= A_2(x'), & \mathcal{P}\psi(x)\mathcal{P}^{-1} &= \sigma^1\psi(x'),\end{aligned}\tag{43}$$

with

$$x = (x_0, x_1, x_2), \quad x' = (x_0, -x_1, x_2),\tag{44}$$

so that the fermion mass term changes sign under parity:

$$\mathcal{P}m\bar{\psi}(x)\psi(x)\mathcal{P}^{-1} = -m\bar{\psi}(x')\psi(x').\tag{45}$$

The same happens with time inversion. Analogous results can be derived for any odd-dimensional fermionic fields.²⁷

Using Eqs. (43) and (44) one can easily show that the Dirac operator Green function changes under parity as follows:

$$\mathcal{P}G_{(m)}(x, y)\mathcal{P}^{-1} = -\sigma^1 G_{(-m)}(x', y')\sigma^1.\tag{46}$$

This formula is meaningful only for $x \neq y$, where the Green function is well defined [we indicate with the subscript (m) the fact we are considering the massive Dirac operator case]. Let us now define the object $\mathcal{J}_{(m)}^\mu(x, y)$:

$$\mathcal{J}_{(m)}^\mu(x, y) = \text{tr } \gamma^\mu G_{(m)}(x, y).\tag{47}$$

Again, this object is well defined whenever $x \neq y$. At $x = y$, precisely where it gives the ground state current [see Eq. (21)], it has to be regulated since, apart from regular terms, it has divergent contributions.

Equation (46) shows that only for $m = 0$ does $\mathcal{J}_{(m)}^\mu(x, y)$ behave as a vector:

$$\begin{aligned}\mathcal{P}\mathcal{J}_{(0)}^0(x, y)\mathcal{P}^{-1} &= \mathcal{J}_{(0)}^0(x', y'), \\ \mathcal{P}\mathcal{J}_{(0)}^1(x, y)\mathcal{P}^{-1} &= -\mathcal{J}_{(0)}^1(x', y'), \\ \mathcal{P}\mathcal{J}_{(0)}^2(x, y)\mathcal{P}^{-1} &= \mathcal{J}_{(0)}^2(x', y').\end{aligned}\tag{48}$$

[Again, the identities (48) make sense only for $x \neq y$.] Then, as argued in I, parity-violating contributions to the ground state current for *massless fermions* cannot

arise from regular terms in $\mathcal{J}_{(0)}^\mu(x, x)$ since these terms should satisfy the transformation law (48). Only additional regular terms generated by the ζ function prescription in the process of giving meaning to $\mathcal{J}_{(m)}^\mu(x, x)$ can break in this case parity invariance: they are not subject to (48) but are just to respect gauge invariance. This is the reason why we were able to compute in I the *complete* parity-violating contribution to $j_\mu(x)$ for arbitrary A_μ : it was not necessary to have a complete knowledge of $G_{(0)}(x, y)$ for arbitrary A_μ but just the behavior of the additional regular terms introduced by the ζ function prescription. One should interpret this parity anomalous contribution present even for massless fermions as a consequence of regularization in odd-dimensional space-time.

It is clear from Eq. (47) that the massive case is more involved: apart from the additional terms generated by the ζ function regularization, regular terms which are already present in $G_{(m)}(x, x)$ [and *a fortiori* in $\mathcal{J}_{(m)}^\mu(x, x)$] may in principle contribute to parity violation since they are not constrained to satisfy vector-like parity transformations. This in turn implies that, in the massive case, one needs a more detailed knowledge of $G_{(m)}(x, y)$. For that reason we shall have to limit the validity of our results to the domain of a perturbative expansion. With this in mind, we rewrite Eq. (20) as

$$j_\mu^{\text{reg}}[A] = -\text{tr} \left[\gamma_\mu \frac{d}{ds} (sK_{s-1}(x, x; D[A])) \right]_{s=0} \quad (49)$$

and use the proposition presented in the previous section in the form given by (40)–(42).

We start analyzing the regular term in Eq. (40) generated by the ζ function method,

$$G^{(\text{local})}(x) = - \int_{|\xi|=1} \frac{d}{ds} C_2(x, \xi; s) \Big|_{s=-1} \frac{d^2 \xi}{(2\pi)^3}. \quad (50)$$

Using the definition for C_2 we have, after differentiation,

$$G^{(\text{local})}(x) = -\frac{i}{2\pi} \int_{|\xi|=1} \left[\int_\Omega \frac{\ln \lambda}{\lambda} [b_{-3}(x, \xi; \lambda)] d\lambda \right] \frac{d^2 \xi}{(2\pi)^3}. \quad (51)$$

As the ξ integral is extended to S_2 , we can see that only terms even in ξ will give nonvanishing contributions to $G^{(\text{local})}(x)$. After some algebra we get from Eq. (28)

$$\begin{aligned} G^{(\text{local})}(x) = & -\frac{i}{2\pi} \int_{|\xi|=1} \frac{d^2 \xi}{(2\pi)^3} \int_\Omega d\lambda \frac{\ln \lambda}{\lambda(\lambda^2 - 1)^3} \\ & \times \{ \lambda [e^2 A^2 - 4e^2 A_\mu A_\nu \xi_\mu \xi_\nu + 3m^2 + 2iem\mathcal{A} \\ & - 8iemA_\mu \xi_\mu \not{x} + ie\partial_\mu A_\nu (-4\xi_\mu \xi_\nu + \delta_{\mu\nu} + i\epsilon_{\mu\nu\alpha} \gamma_\alpha)] \\ & + \lambda^3 [-e^2 A^2 - 2iem\mathcal{A} + m^2 - ie\partial_\mu A_\nu (\delta_{\mu\nu} + i\epsilon_{\mu\nu\alpha} \gamma_\alpha)] \}. \quad (52) \end{aligned}$$

At this point we have to choose a curve Ω in order to perform the λ integral, this choice determining the branch for the log function in Eq. (52). Let us first consider the curve $\Omega_{(+)}$, depicted in Fig. 1, and call

$$I_{(+)}[p, q] = \int_{\Omega_{(+)}} \frac{\lambda^q \ln \lambda}{(\lambda^2 - 1)^p} d\lambda. \quad (53)$$

It is easy to see that the integration along the small circle around the origin vanishes for $q \geq 0$ as its radius goes to zero while the integrals along the ray sum up to give

$$I_{(+)}[p, q] = i^q (-1)^p \pi B\left(\frac{q+1}{2}, p - \frac{q+1}{2}\right), \quad (54)$$

where $B(m, n)$ is the β function (Euler's integral of the first kind),²⁸ defined as

$$B(m, n) = \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}} dt. \quad (55)$$

The β functions can be evaluated in terms of Γ functions to give

$$I_{(+)}[p, q] = i^q (-1)^p \pi \frac{\Gamma(\frac{q+1}{2}) \Gamma(p - \frac{q+1}{2})}{\Gamma(p)}. \quad (56)$$

Using this result and performing the remaining ξ integral we get

$$G_{(+)}^{(\text{local})}(x) = -\frac{ie}{8\pi} \epsilon_{\mu\nu\alpha} \partial_\mu A_\nu \gamma_\alpha + \frac{i}{4\pi} m^2, \quad (57)$$

where the subscript $(+)$ indicates that we have used the $\Omega_{(+)}$ contour. The corresponding contribution to the fermionic current (49) is

$$j_{\mu(+)}^{(\text{local})} = -\text{tr}[\gamma_\mu G_{(+)}^{(\text{local})}(x)] = \frac{ie}{4\pi} \epsilon_{\alpha\beta\mu} \partial_\alpha A_\beta. \quad (58)$$

If instead we choose the curve Ω_- (see Fig. 1), we get the same expression but with the opposite sign, so we finally have, for the regular term generated by the ζ function method,

$$j_{\mu\pm}^{(\text{local})} = \pm \frac{ie}{4\pi} \epsilon_{\alpha\beta\mu} \partial_\alpha A_\beta. \quad (59)$$

Let us first discuss in more detail the origin of the sign ambiguity in (59). As discussed in I, $I_\Omega[p, q]$ in Eq. (53) satisfies

$$I_{\Omega_{(-)}}[p, q] = (-1)^{q+1} I_{\Omega_{(+)}}[p, q], \quad (60)$$

where $\Omega_{(+)}$ is any curve that avoids a ray of minimal growth of $D[A]$ on the upper half-plane and $\Omega_{(-)}$ is any other curve that avoids such a ray on the lower half-plane. Now, for odd-dimensional spaces, only even values for q arise. In the present $d = 3$ case, $q = 0, 2$. On the other hand, for even-dimensional spaces, $q = 2k + 1$.

This is the reason why there are sign ambiguities in computing anomalies in odd-dimensional spaces (according to the choice of curves Ω), which are not present for even dimensions. Note that Seeley's definition of complex powers of elliptic operators does not exclude the possibility of a multivaluation, for example in the form of a double sign.

Another important lesson learnt in this calculation is that this result is identical to that corresponding to massless fermions. Indeed, *no new terms* arise from $G_{(\pm)}^{(\text{local})}$ due to the presence of the mass term since, as happens when one is computing the chiral anomaly,²¹ the trace in (49) makes all the terms proportional to m and m^2 vanish.

We thus see that the contribution to parity violation in $j_\mu^{\text{reg}}[A]$ arising from the regular term generated by the ζ function method is independent of whether the fermions are massless or massive (even if the mass is a regulating mass that should be taken to zero at the end of the calculation). Its origin can be traced back to gauge-invariant regularization in odd-dimensional spaces and it is present in both the massless and the massive case.

We now have to consider regular terms coming from the brackets in Eq. (35), i.e. the regular part of $G_{(m)}(x, x)$ remaining once the adequate subtractions are performed. As stated above, there was no contribution to parity violation from these terms in the massless case due to the transformation law (46) obeyed by $G_{(0)}(x, x)$: parity was respected at the classical level so that violations cannot arise in the process of subtracting singular terms to $G_{(0)}(x, x)$. On the contrary, there may be parity-violating contributions in the case of the massive Green function since already at the classical level parity is violated by the massive Dirac operator. Should these contributions exist, they will be the product of parity noninvariance introduced in the Lagrangian through the fermionic mass term. To see whether this happens, we are faced with the necessity of making some sort of approximation to compute $G_{(m)}(x, y)$ at $x \neq y$ and, from it, the regular part of $G_{(m)}(x, x)$ defined by $G^{(\text{subt})}(x)$ in Eq. (41).

Of course, for $A_\mu = 0$ the Green function can be computed closely:

$$G_{(m)}(x, y; A_\mu = 0) = \frac{1}{4\pi} (i\not{\partial} - im) \frac{e^{-|m||x-y|}}{|x-y|}. \quad (61)$$

It is then natural to seek for a perturbative expansion in powers of the coupling constant e :

$$\begin{aligned} G_{(m)}(x, y; A_\mu) &= G_{(m)}(x, y; A_\mu = 0) \\ &\quad - e \int d^3w G_{(m)}(x, w; A_\mu = 0) \not{A}(w) G_{(m)}(w, y; A_\mu = 0) + O(e^2). \end{aligned} \quad (62)$$

We will keep only the first order in the perturbative expansion, but a systematic perturbative calculation can be straightforwardly implemented. In order to analyze an arbitrary gauge field configuration, we will make a Taylor expansion of $A_\mu(w)$ around x . This leads to a derivative expansion which for dimensional analysis takes the form of a ∂/m expansion.²⁹ Up to first derivatives the Green function can be expressed as a Laurent expansion in $z_\mu = y_\mu - x_\mu$ as

$$\begin{aligned} G_{(m)}(x, y; A_\mu) = & \frac{1}{4\pi} \left[i \frac{\not{z}}{|z|^3} - im \frac{1}{|z|} - \frac{im^2}{2} \frac{\not{z}}{|z|} + im|m| \right. \\ & + eA_\alpha(x) \frac{z_\alpha \not{z}}{|z|^3} - emA_\alpha(x) \frac{z_\alpha}{|z|} - \frac{ie}{2} \frac{m}{|m|} \epsilon_{\mu\nu\alpha} \partial_\mu A_\nu(x) \gamma_\alpha \\ & \left. + \frac{ie}{2} \epsilon_{\mu\nu\alpha} \partial_\mu A_\nu(x) \frac{z_\alpha}{|z|} + \frac{e}{2} \partial_\mu A_\nu(x) \frac{z_\mu z_\nu \not{z}}{|z|^3} \right] + O(z). \end{aligned} \quad (63)$$

As expected, nonregular terms do appear on the diagonal of the Green function ($z_\mu \rightarrow 0$). According to proposition (35) these terms should be canceled by the subtractions. Indeed, the explicit values of these subtractions are

$$\int \frac{d^3\xi}{(2\pi)^3} C_0(x, \xi/|\xi|; -1) |\xi|^{-1} e^{i\xi \cdot (x-y)} = \frac{1}{4\pi} i \frac{\not{z}}{|z|^3}, \quad (64)$$

$$\int \frac{d^3\xi}{(2\pi)^3} C_1(x, \xi/|\xi|; -1) |\xi|^{-2} e^{i\xi \cdot (x-y)} = \frac{1}{4\pi} \left[-im \frac{1}{|z|} + eA_\alpha(x) \frac{z_\alpha \not{z}}{|z|^3} \right], \quad (65)$$

$$\begin{aligned} & \int_{|\xi| \geq 1} \frac{d^3\xi}{(2\pi)^3} C_2(x, \xi; -1) e^{i\xi \cdot (x-y)} \\ & = \frac{1}{4\pi} \left[-\frac{im^2}{2} \frac{\not{z}}{|z|} - emA_\alpha(x) \frac{z_\alpha}{|z|} \right. \\ & \quad \left. + \frac{ie}{2} \epsilon_{\mu\nu\alpha} \partial_\mu A_\nu(x) \frac{z_\alpha}{|z|} + \frac{e}{2} \partial_\mu A_\nu(x) \frac{z_\mu z_\nu \not{z}}{|z|^3} \right] + O(e^2), \end{aligned} \quad (66)$$

so that by the use of (35) nonregular terms are exactly canceled. In the last expression we used Eqs. (36)–(38), where $M(x) = 0$; by $O(e^2)$ we just mean some nonregular terms proportional to e^2 . One can check that each subtraction corresponds to nonregular terms that are homogeneous of degree -2 , -1 and 0 respectively in z_μ .

The regular part of $G_{(m)}(x, x)$ remaining after subtractions and the limit $y \rightarrow x$ are performed is

$$G^{(\text{subt})}(x) = \frac{1}{4\pi} \left[im|m| - \frac{ie}{2} \frac{m}{|m|} \epsilon_{\mu\nu\alpha} \partial_\mu A_\nu(x) \gamma_\alpha \right]. \quad (67)$$

The corresponding contribution to the current reads

$$j_{\mu}^{(\text{subt})}[A] = -\text{tr}[\gamma_{\mu} G^{(\text{subt})}(x)] = \frac{ie}{4\pi} \frac{m}{|m|} \epsilon_{\mu\nu\alpha} \partial_{\nu} A_{\alpha}(x). \quad (68)$$

The complete expression for the the fermionic current (up to order e and up to first derivatives of the gauge field) can now be obtained by adding $j_{\mu}^{(\text{local})}$ in Eq. (59) and $j_{\mu}^{(\text{subt})}$ in Eq. (68). The answer reads

$$j_{\mu}^{\text{reg}} = \frac{ie}{4\pi} \left(\pm 1 + \frac{m}{|m|} \right) \epsilon_{\mu\nu\alpha} \partial_{\nu} A_{\alpha}(x). \quad (69)$$

In order to compare this result with those obtained in previous works, let us first briefly insist on its domain of validity. Original works showing parity violation and the consequent emergence of a CS term for massless and massive fermions were one-loop calculations performed in a constant $F_{\mu\nu}$ background. For massless fermions we were able in I, through the ζ function approach, to evaluate the *complete* parity violation contribution for an *arbitrary* $F_{\mu\nu}$ background. This contribution reappears now in the massive case and corresponds to the first term of Eq. (69). Again this contribution corresponds to an *exact* result (not just a one-loop contribution) valid for *arbitrary* A_{μ} . Now, in the massive case one has also to consider the contribution given by $G^{(\text{subt})}(x)$, i.e. the parity violation coming from the fermion mass term. This we were not able to evaluate in a closed form due to the impossibility of calculating the Dirac Green function in an arbitrary background. For that reason, we had to appeal to some sort of approximation. We chose a perturbative calculation still valid for arbitrary $F_{\mu\nu}$ and we stopped at the first order in e . We also neglected $O(\partial^2/m^2)$ in a derivative expansion. Of course, our perturbative expansion can be systematically employed to calculate higher order contributions to the second term of Eq. (67). Although we made the explicit calculations for Abelian gauge fields, the ζ function approach can be used with no further modifications in the non-Abelian case.

Equation (69) is the main result in our paper and we shall devote the next section to the discussion of its origin and implications.

4. Discussion

Through the presence of a term proportional to the Levi–Civita pseudotensor in the ground state fermionic current, Eq. (69) shows that parity is violated when massive fermions are considered in three-dimensional space–time. This in turn produces a CS term in the effective action for the gauge field.

A first important point to note in the formula (69) concerns the existence of two clearly differentiated sources of parity violation. One is related with the fact that a mass term necessarily violates P and T in three dimensions. This fact leads to the second term of (69). In contrast, the first term of (69) originated in peculiarities

of regularization in odd-dimensional space-times and has nothing to do with the fermion mass. It is a pure ultraviolet effect (it arises from the necessity of making regular the singular behavior of the Green function at $x = y$) while the second term of (69) can be interpreted as an infrared contribution which cannot be naively extended to the $m = 0$ case. In fact, we proved in I that this term is absent if one starts with $m = 0$.

Once we get the ζ function answer to the parity violation term in the fermionic current, it is natural to try to understand the disagreement of the corresponding result with those obtained in Refs. 6 and 7. First, we note that for $m = 0$ there is no disagreement at all, except for the interpretation of the origin of the anomaly. Indeed, quantitatively, the answer in Refs. 6 and 7 and in I was the same. Now, while the double sign origin was traced back in the former references to the necessity of introducing regulating mass, it was clear in the ζ function approach, where no mass term is introduced by hand, that it arises from properties of odd-dimensional spaces.

When the fermions are massive, there are quantitative discrepancies apart from interpretational differences. Indeed, the ζ function answer contains an additional contribution not taken into account for example in Ref. 7. This situation much resembles what happens in even-dimensional spaces concerning gauge anomalies: also in that case different regularization schemes lead to two different results for the anomaly in the gauge current conservation equation. One is known as the covariant anomaly, and the other one as the consistent anomaly (the ζ function approach leading automatically to the covariant result; a particular heat kernel approach leading for example to the consistent one²¹). As explained in Ref. 30, it is on physical grounds that one should decide which result to use or, in other words, which regularization scheme one has to adopt.

Keeping this analogy in mind, we conclude that the particular physical situation will determine which result one should use for the parity anomaly. We think that our treatment has shown, however, the unnaturalness of those proposals in which, starting with an even number of fermions and then choosing half the masses with one sign, half with the other, parity conservation is achieved: there is a parity-violating effect which is intrinsic to odd dimensions and which cannot be accommodated, in our opinion, by a clever choice of mass signs. It is there to remain and it should serve as a guide even when other regularization prescriptions are adopted.

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